

Predicting the chemical output of the human gut microbiome

A machine-learning system that infers a gut metabolomic profile from bacterial composition alone, evaluated honestly enough to say where it fails.

Phase 2 report – data expansion, outcome layer, adversarial backtesting, and public interface

Status Research prototype – not a medical device

Data 2,900 paired samples · 1,759 subjects · 14 cohorts

Model 1,098 metabolites · 5,038 taxonomic features

Date 13 September 2026

Principal finding. The model achieves a median held-out correlation of $r = 0.458$ on patients from cohorts it trained on, and approximately zero ($r = 0.002$) on cohorts it has never seen. Reported disease-classification performance in this domain is shown to be largely an artefact of batch confounding. Both results are reproducible from this repository.

What was built, and what it showed

7448 AI predicts which chemicals a gut bacterial community will produce, using only a sequencing profile of that community. The motivation is an asymmetry in available data: sequencing is cheap and ubiquitous, metabolomics is expensive and rare, so a reliable mapping from one to the other would give a chemical readout to thousands of existing datasets that never paid for one.

This phase extended an existing baseline in four directions: substantially more reference data, a layer connecting predictions to documented health outcomes, a set of adversarial backtests, and a public-facing application that exposes the model's reasoning rather than concealing it. The engineering outcomes were successful. The scientific outcome is more equivocal, and that equivocation is the most valuable result reported here.

0.458 median held-out correlation, within cohorts	0.002 median correlation on an unseen cohort	0.643 disease AUROC once batch is controlled	7,163 sourced microbe-disease associations added
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Three findings that matter

- 1. Within-cohort accuracy is real.** A shuffled-label negative control scored $r = -0.002$ against $r = 0.599$ on the same metabolites with real labels. The pipeline does not leak its targets into its inputs, so the signal it finds inside the training distribution is genuine.
- 2. The model does not transfer between laboratories.** Holding out an entire cohort and predicting it from the other thirteen produces a median correlation of 0.002, with the best of fourteen cohorts reaching only 0.126. Because Pearson correlation is invariant to rescaling, this cannot be dismissed as a units mismatch between mass-spectrometry platforms. It is a genuine failure to rank samples correctly in an unfamiliar dataset, and it is the project's central open problem.
- 3. Published-style disease classification is mostly batch effect.** A classifier pooling all cohorts reaches AUROC 0.958. A classifier given no biological information at all — only which cohort produced a sample — reaches 0.915. Restricting to case-versus-control within a single cohort, where laboratory and platform are held constant, the honest figure is 0.643, and only 6 of 17 condition×cohort models exceed chance at a 95% confidence interval.

WHY THIS IS REPORTED PROMINENTLY RATHER THAN BURIED

A tool that discusses human health earns trust by being falsifiable. The commercial incentive in this field is to publish the 0.958 and omit the control that explains it; doing so produces products that appear validated and are not. Every number above is regenerated from committed experiment outputs by a single command, and the public interface displays the unflattering ones on a dedicated page.

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Background

The human large intestine hosts on the order of 10^{13} bacteria. They are metabolically active: they ferment dietary components the host cannot digest and, in doing so, synthesise hundreds of small molecules that are absorbed into circulation and act on host tissue.

1.1 Why the chemistry matters more than the census

Most microbiome research reports *who is present*. The more consequential question is usually *what they are producing*. Butyrate, a short-chain fatty acid generated by fermentation of dietary fibre, supplies the majority of the energy used by colonocytes — the cells lining the colon take their fuel directly from bacterial output rather than from circulating glucose. Trimethylamine-N-oxide (TMAO) is produced in two steps, a bacterial conversion of dietary choline and carnitine followed by hepatic oxidation, and circulating levels have been associated with cardiovascular risk across multiple cohorts. Secondary bile acids exist only because a restricted set of gut bacteria chemically modify host-synthesised primary bile acids.

In each case the biologically active quantity is a metabolite, not a taxon. Two communities with similar membership can differ substantially in output, and community composition is an imperfect proxy for function.

1.2 The data asymmetry

Both quantities are measurable. 16S rRNA and shotgun metagenomic sequencing are inexpensive, standardised, and performed at scale; untargeted metabolomics by mass spectrometry is costly, slower, and available in comparatively few laboratories. The consequence is that the overwhelming majority of the world's microbiome data records composition without chemistry.

If composition predicts chemistry with characterised accuracy, that accumulated data acquires a chemical dimension retrospectively, at no additional experimental cost. That is the premise of this project. The qualifier *with characterised accuracy* is load-bearing, and most of this report concerns establishing what that accuracy actually is.

SCOPE AND INTENT

This is a research instrument for understanding relationships between microbial composition and gut chemistry. It is not a diagnostic device, has no regulatory clearance, and is not validated for clinical use on individual samples. The accompanying application is explicit about this on every page that displays a health-related output.

Prior art and positioning

Predicting metabolites from microbiome composition is an established problem with established baselines. The contribution of this phase is not a novel architecture — it is the evaluation.

2.1 Existing approaches

MelonnPan (Mallick et al., 2019) fits an independent elastic-net regression per metabolite from microbial feature abundances, and remains the standard reference point. **MIMOSA2** takes a mechanistic rather than statistical route, using genome-scale metabolic reconstruction to predict which reactions a community can perform. Subsequent work has applied random forests, gradient boosting and neural architectures to the same task.

The recurring methodological difficulty across this literature is sample size relative to dimensionality. Paired microbiome–metabolome cohorts are small — typically tens to a few hundred subjects — while feature spaces run to thousands of taxa and outputs to hundreds or thousands of metabolites. This is an unfavourable regime for high-capacity models, and the project's own results reproduce that conclusion.

2.2 What this project adds

- **An adversarial evaluation suite.** A shuffled-label leakage control, leave-one-cohort-out transfer testing, split-stability testing across seeds, and a batch-identity control for the disease classifier. These are reported whether or not they flatter the model.
- **A sourced outcome layer.** 7,163 curated microbe–disease associations drawn from 1,179 publications, each retaining its citation, the number of supporting reports, the consistency of their direction, and per-study methodological quality flags.
- **Exact interpretability.** Because the deployed model is linear, each prediction decomposes precisely into per-microbe contributions that sum to the output. No post-hoc attribution method is used, because none is needed.
- **A public interface built around uncertainty.** Every predicted value is displayed with the held-out accuracy achieved for that specific metabolite.

Data

3.1 Training corpus

The model trains on the Borenstein Lab curated microbiome–metabolome collection: fourteen independent published cohorts in which the same stool samples were both sequenced and analysed by mass spectrometry, harmonised by the original curators.

Table 1 — Training corpus after standardisation

PROPERTY	VALUE	NOTES
Paired samples	2,900	Sequencing and metabolomics on the same sample
Distinct subjects	1,759	Six cohorts sampled individuals repeatedly
Independent cohorts	14	Separate studies, laboratories and platforms
Genus-level features	12,263	5,038 retained after a 5% prevalence filter
Metabolite targets	1,384	1,098 with sufficient coverage to model
Training samples	2,468	Train + validation; test held out entirely

Two properties of this corpus shaped the pipeline. First, **taxonomic resolution varies**: only six of fourteen cohorts provide species-level profiles, so genus is the common feature space. Second, and more consequentially, **a missing metabolite value does not mean zero** — cohorts ran different metabolomics panels, so a blank usually means the compound was not on that panel. These are tracked by an explicit boolean mask, and each metabolite trains only on samples where it was genuinely measured. Treating blanks as zeros would teach the model to predict absence for any compound that happened to be unfashionable to assay.

3.2 Sources added in this phase

Table 2 — External data acquired, all through documented public interfaces

SOURCE	CONTRIBUTION	SCALE
Disbiome	Curated observations of microbes reported elevated or reduced in a condition, with citations and per-study quality flags	10,866 records → 7,163 associations
KEGG	Compound pathway membership and BRITE chemical classification	967 of 1,384 targets annotated
PubMed	Citation resolution for every association displayed	1,179 papers median year 2018
Cohort metadata	Longitudinal structure recovered from per-cohort time columns	331 subjects 1,372 samples

Of the 7,163 genus–disease associations, 5,610 fall on genera present in the model's feature space and 3,958 are supported by at least one study of a gut or stool sample specifically. The distinction matters: a genus

reported elevated in saliva during periodontitis carries no implication for a stool profile, so sample site is tracked per record rather than pooled.

TWO SOURCES DELIBERATELY NOT USED

HMDB holds disease annotations for exactly the identifiers this project keys on, but `hmdb.ca` returns HTTP 403 to scripted requests. Rather than circumvent that control, KEGG was used for equivalent enrichment through its documented API. **GMrepo** would have supplied additional phenotype-labelled samples, but every endpoint tested returned a server-side `RuntimeError`. Both outcomes are recorded in the repository rather than worked around; data acquired by defeating a publisher's access controls would also be data of unverifiable provenance.

Methods

4.1 Metabolite prediction

The deployed model is an elastic-net regression per metabolite — 1,098 independent models — following the MelonnPan formulation. Per-target regularisation parameters were selected by cross-validated search on training data only. Features are genus-level relative abundances under an arcsine-square-root transform, the standard variance-stabilising choice for compositional data.

A deep alternative was evaluated on identical splits: a four-layer tapering feed-forward network (3000–2000–1000–500) with batch normalisation and dropout, predicting all metabolites jointly under a masked loss. It reached a median held-out correlation of 0.376 against the elastic net's 0.458. With roughly 2,000 training samples against 5,038 features, the data volume does not support the additional capacity, and the simpler model was deployed.

A USEFUL CONSEQUENCE OF THE LINEAR MODEL WINNING

For a linear model the contribution of feature j to prediction i is exactly coefficient $\times$ input, and these contributions sum to the prediction. The "which microbes produced this number" view in the application is therefore not an approximation of the model's reasoning — it is the model's reasoning, read directly from the weights. Attribution methods such as SHAP exist to estimate this where it cannot be read off; using one here would add error to recover a quantity already available exactly.

4.2 Evaluation design

Splits are by individual, not by sample. Six cohorts sampled the same person repeatedly; a sample-level split would place one individual's samples on both sides of the boundary and reward the model for recognising the person. All splits assign whole subjects, stratified by cohort.

Four backtests were implemented beyond the standard held-out split:

Table 3 — Backtest suite

TEST	QUESTION	DESIGN
Negative control	Does the pipeline leak targets into features?	Permute each metabolite's values across training subjects, refit unchanged. Correct pipelines score zero.
Cross-cohort	Does it transfer to an unseen laboratory?	Train on 13 cohorts, predict the 14th; repeat for all 14.
Split stability	Is the headline number an artefact of one split?	Redraw the subject-level split under multiple seeds and refit.
Chemical class	Is accuracy uniform across compound types?	Stratify held-out accuracy by KEGG BRITE classification.

4.3 The outcome layer

Cohort disease labels were harmonised into a single vocabulary — six distinct spellings of "healthy control" appear across the fourteen studies — producing 1,541 case and 887 control samples across 14 conditions.

Twelve cohorts contain both case and control arms and therefore support within-cohort comparison.

Three classifiers were trained per condition, deliberately, because only their comparison is interpretable: a within-cohort model (cases versus controls from one laboratory), a pooled model (all cohorts together), and a control model given *only* a one-hot encoding of cohort identity and no biological features whatsoever. Section 6 reports what that comparison revealed.

4.4 Forward outlook

Longitudinal structure was recovered from six cohorts, yielding 331 subjects with two or more timepoints spanning up to 967 days. No forecasting model was fitted. With this sample size, across six studies of different populations sampled on different schedules, a projected trajectory would imply precision the data cannot support. Instead the system performs an empirical analogue lookup: it identifies subjects whose starting composition most resembles a query profile and reports the observed distribution of what was subsequently measured in them, with sample size attached.

Results

5.1 Metabolite prediction

0.458 median held-out r	418 metabolites at $r \geq 0.5$	621 metabolites at $r \geq 0.3$	185 metabolites below $r = 0.2$
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Accuracy is strongly heterogeneous across metabolites, which is why the application displays a per-metabolite accuracy figure beside every value rather than a single headline number. Several compounds of established biological interest are predicted well.

Table 4 — Selected well-predicted metabolites of biological interest

METABOLITE	RELEVANCE	HELD-OUT R
Glycochenodeoxycholate	Bile acid metabolism	0.95
Imidazole propionate	Associated with insulin resistance	0.91
Lithocholate	Bacterially-produced secondary bile acid	0.92
5-Hydroxytryptophan	Serotonin precursor	0.90
TMAO	Cardiovascular risk marker	0.86
Indole-3-propionate	Gut barrier function	0.84
Hippurate	Marker of microbial diversity	0.83
Acetate	Most abundant short-chain fatty acid	0.72
Butyrate	Primary colonocyte energy source	0.69
Propionate	Hepatic glucose metabolism	0.57

5.2 Leakage control

RESULT: PASS

Across 60 densely-measured metabolites, models fitted on real data scored a median $r = 0.599$. The same models fitted on permuted labels scored $r = -0.002$. The within-distribution signal is real and the pipeline does not leak its targets into its inputs.

5.3 Disease classification

Six of seventeen condition×cohort models exceed chance once a bootstrap 95% confidence interval is applied to held-out patients. The remainder do not, and are treated as unvalidated throughout the application.

Table 5 — Within-cohort models exceeding chance (bootstrap 95% CI excludes 0.5)

CONDITION	COHORT	AUROC	95% CI	N
Crohn's disease	Franzosa 2019	0.986	0.94–1.00	24
Neonatal sepsis	Wandro 2018	0.958	0.78–1.00	11
End-stage renal disease	Wang 2020	0.893	0.73–1.00	43
Ulcerative colitis	Franzosa 2019	0.889	0.70–1.00	20
IBS (constipation)	Mars 2020	0.871	0.73–0.98	32
IBS (diarrhoea)	Mars 2020	0.730	0.56–0.89	61

Test-set sizes are small, and the confidence intervals are correspondingly wide. Several of these results rest on tens of patients. They are reported with intervals precisely so that they are not mistaken for settled findings.

5.4 Association layer coverage

The literature layer reproduces well-established findings, which serves as a sanity check on the ingestion pipeline: *Faecalibacterium* reduced in Crohn's disease (30 reports, 100% directional agreement), *Fusobacterium* elevated in colorectal cancer (33 reports, 94% agreement), and *Roseburia* reduced in Crohn's disease (17 reports, 100% agreement). These are among the most replicated results in the field, and their correct recovery indicates the association layer is functioning as intended.

The generalisation failure

This section reports the most important result of the phase. It is a negative one, and it qualifies every number in Section 5.

6.1 Cross-cohort transfer is absent

The standard evaluation — held-out patients drawn from the same fourteen cohorts used for training — yields a median correlation of 0.458. Holding out an entire cohort instead, and predicting it from the remaining thirteen, yields a median correlation of **0.002**.

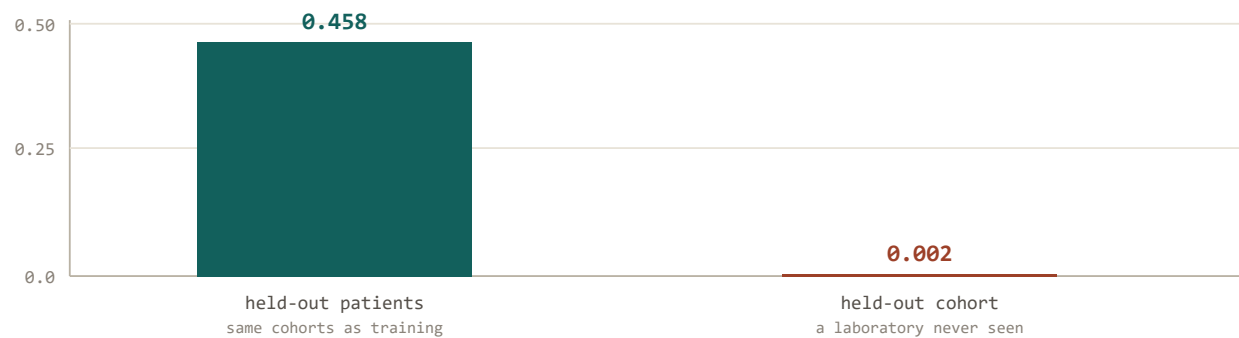


Figure 1 — Median held-out Pearson correlation under the two evaluation regimes. The left bar is the figure normally reported for models of this type. The right bar is the same model evaluated on a cohort withheld entirely from training, across all 14 cohorts, on 60 densely-measured metabolites.

Across the fourteen held-out cohorts the best median correlation was 0.126 and the worst was -0.162 . A natural objection is that different mass-spectrometry platforms report different units, so cross-cohort prediction should fail on scale alone. That objection does not hold: Pearson correlation is invariant under linear rescaling, so a pure units mismatch would not reduce it. The model is failing to rank samples correctly within the unseen cohort — a substantive failure, not a calibration artefact.

6.2 The batch-confounding control

An analogous problem affects disease classification, and here a direct control quantifies it. Each cohort in this collection studied one condition, on one sequencing platform, in one laboratory. Disease status is therefore almost perfectly correlated with cohort identity, and a classifier can score well by recognising a laboratory's technical signature rather than any biology.

To measure how much of the apparent performance this accounts for, a third classifier was trained on a one-hot encoding of cohort identity and nothing else — no taxonomic features at all.

Table 6 — Disease classification under three designs

DESIGN	INFORMATION AVAILABLE TO THE MODEL	MEDIAN AUROC
Pooled across cohorts	Taxonomy, plus cohort identity implicitly	0.958
Cohort identity only	Which laboratory produced the sample. No biology.	0.915
Within cohort	Taxonomy only; laboratory held constant	0.643

INTERPRETATION

A model containing no biological information whatsoever reaches AUROC 0.915, against 0.958 for the model that also sees the microbiome. Nearly all of the apparent predictive performance in the pooled design is attributable to batch signature. The defensible estimate of biological signal is the within-cohort figure of 0.643, obtained where laboratory and platform are constant by construction.

6.3 Consequences for the project

These two results are consistent with one another and point to the same conclusion: the system has learned relationships that hold *within* this collection of datasets, and has not demonstrated relationships that hold across laboratories. That is a meaningful scientific object — the within-cohort signal survives a stringent leakage control — but it is not yet a deployable instrument for an arbitrary new sample.

Accordingly, the roadmap in Section 9 was reordered. Architectural refinement aimed at incremental within-cohort accuracy has been deprioritised below work targeting transfer directly, since improvements to a figure that does not generalise have limited value.

The application

A web interface makes the model usable by someone with no background in microbiology, and — equally importantly — makes its reasoning and its limitations inspectable.

7.1 Structure

Table 7 — Interface sections

SECTION	FUNCTION
Overview	Plain-language explanation of the microbiome, metabolites, and the premise, assuming no prior knowledge
Simulator	Adjust bacterial abundances and observe predicted chemical changes live, with sourced associations and a population map
Outlook	Empirical analogue lookup over longitudinal subjects
Inside the model	The batch-confounding control, the transfer failure, the leakage check, and the complete pipeline run log
Methodology	Data provenance, evaluation design, limitations, privacy posture

7.2 Design commitments

- **No value without its uncertainty.** Every predicted metabolite is rendered with the held-out correlation achieved for that specific compound. A metabolite at $r = 0.12$ is visually distinguished from one at $r = 0.69$ rather than presented identically.
- **Change is the headline, not level.** The simulator leads with the difference from a reference profile, because systematic model error largely cancels in a difference while an absolute value carries it in full.
- **Adjustments apply over a real population profile.** A user adjusting five taxa has said nothing about the other 5,033; zeroing them would submit a composition no human gut has ever had.
- **Associations carry citations.** Each displayed association shows supporting study count, directional consistency, study quality, recency, and links to the papers.
- **Unflattering results are a page, not a footnote.** The transparency section leads with the batch-effect control and the transfer failure, generated from committed experiment outputs rather than written by hand.

7.3 Privacy and security posture

The service is stateless by design. It holds no database, sets no cookies, creates no accounts, and does not log submitted compositions; each response is a pure function of the request and a fixed model snapshot. For a tool discussing health-adjacent data, storing nothing is the strongest available privacy guarantee, and it costs nothing here. The model is fixed and does not learn from traffic, which is what makes the transparency view meaningful: the model a visitor inspects is the model that answered them.

Controls implemented: schema validation with explicit bounds on every input, request body size limits, per-client rate limiting, a restrictive content security policy and associated headers, environment-configured CORS with no wildcard, and containers running unprivileged and excluding both the training data and the

training code. The application carries a visible non-diagnostic disclaimer on every page displaying health-related output.

Limitations

8.1 Generalisation

The system has not demonstrated transfer to an unseen laboratory (Section 6). All reported accuracy should be read as within-distribution. This is the binding constraint on the project.

8.2 Batch confounding

Cohort identity is nearly collinear with disease status throughout this collection. Within-cohort analysis mitigates but does not eliminate the problem, and it restricts the usable sample size to one cohort at a time.

8.3 Composition without function

The model observes which bacteria are present, not which genes they express. Gene- and pathway-level profiling (HUMAnN3) is reported in the literature to be more predictive than taxonomy alone and is absent here; generating it requires raw sequence reads and a Linux bioinformatics toolchain, making it infrastructure work rather than a modelling change.

8.4 Taxonomic resolution

The common feature space stops at genus, because only six of fourteen cohorts provide species-level profiles. Biologically important distinctions exist below that level; *E. coli* strains range from commensal to pathogenic.

8.5 Association, not causation; group, not individual

The outcome layer reports statistical associations observed in populations. Direction of causality is generally unestablished, and many associations are plausibly downstream of diet or medication rather than upstream of disease. Nothing in this system supports inference about an individual.

8.6 Representativeness

The source cohorts skew toward particular geographies, age ranges and dietary patterns. A profile from an under-represented population is compared against a reference that describes it poorly.

8.7 Statistical power

Several validated disease models rest on tens of held-out patients, with correspondingly wide confidence intervals. The longitudinal analysis draws on 331 subjects across six heterogeneous studies.

Roadmap

Reordered by the Phase 2 findings. There is limited value in improving a within-cohort figure that does not transfer, so generalisation work is promoted above architectural refinement.

Priority 1 — Close the generalisation gap

Four approaches are testable with the existing harness, in approximate order of expected value:

- **Per-cohort target normalisation.** Rank-normalise metabolite values within cohort before training, so the model learns ordering rather than platform-specific scale.
- **Explicit batch correction.** ComBat or MMUPHin applied to features prior to modelling.
- **Domain-adversarial training.** Penalise representations from which cohort identity is recoverable, directly targeting the confound Section 6 quantifies.
- **Leave-one-cohort-out model selection.** Select hyperparameters against transfer performance rather than random-split performance, so the selection criterion matches the deployment condition.

Success criterion: median cross-cohort correlation materially above zero. Until that is achieved, other improvements are secondary.

Priority 2 — Functional profiling

Gene and pathway abundance via MetaPhlAn and HUMAnN3. Beyond the expected accuracy gain, there is a specific reason to expect it to help with transfer: gene content is plausibly less laboratory-specific than 16S composition, since it is less sensitive to primer choice and amplification protocol. Requires raw FASTQ retrieval and a Linux or WSL bioconda environment.

Priority 3 — Additional labelled cohorts

curatedMetagenomicData is the most substantial available source. GMrepo was attempted and is currently returning server errors. Additional cohorts serve the generalisation problem directly by increasing the diversity of laboratories represented.

Priority 4 — Architecture

Encoder–decoder bottleneck and multi-task formulations, which share statistical strength across correlated metabolites more efficiently than a wide feed-forward network. To be evaluated on cross-cohort transfer, not only on held-out patients.

Priority 5 — Strain resolution and mechanism

Species- or strain-level features where cohorts support them, and integration of mechanistic constraints in the MIMOSA2 tradition — restricting predictions to reactions the observed community plausibly encodes.

Reproducibility

Every figure in this report regenerates from the repository. Each pipeline step appends a structured record — parameters, metrics, output artefacts, duration, and failures — to `experiments/run_log.jsonl`, which the application also reads and displays. `python scripts/status.py` prints current pipeline state.

Table 8 — Regenerating the principal results

RESULT	COMMAND
Median held-out $r = 0.458$	<code>scripts/train_elastic_net.py</code>
Elastic net vs. MLP	<code>scripts/compare_baselines.py</code>
Cross-cohort $r = 0.002$	<code>scripts/backtest.py --test cross_cohort</code>
Leakage control	<code>scripts/backtest.py --test negative_control</code>
Batch-identity control	<code>scripts/train_outcome_model.py</code>
Association layer	<code>scripts/fetch_disbiome.py</code>
Deployable model	<code>scripts/train_production_model.py</code>

A test suite of 46 tests covers lineage parsing, inference and attribution correctness — including an assertion that per-feature contributions sum to the prediction, the property the interface's explanation view depends on — and API contract and input validation.

References and data sources

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STATUS DECLARATION

7448 AI is a research prototype. It is not a medical device, has not been evaluated by any regulatory body, and must not be used to diagnose, treat, or make decisions about any individual's health. Its principal validated finding to date is a negative one: that predictive performance of this class of model, evaluated conventionally, substantially overstates what it achieves on data from an unfamiliar source.